

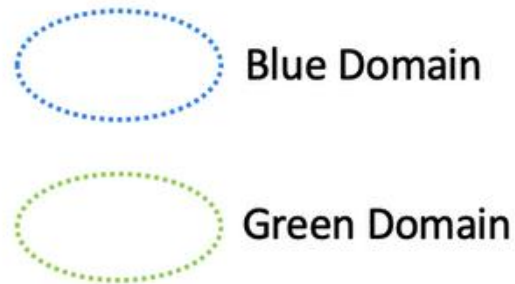


Network Segmentation

ACE Team

Network Segmentation - Overview

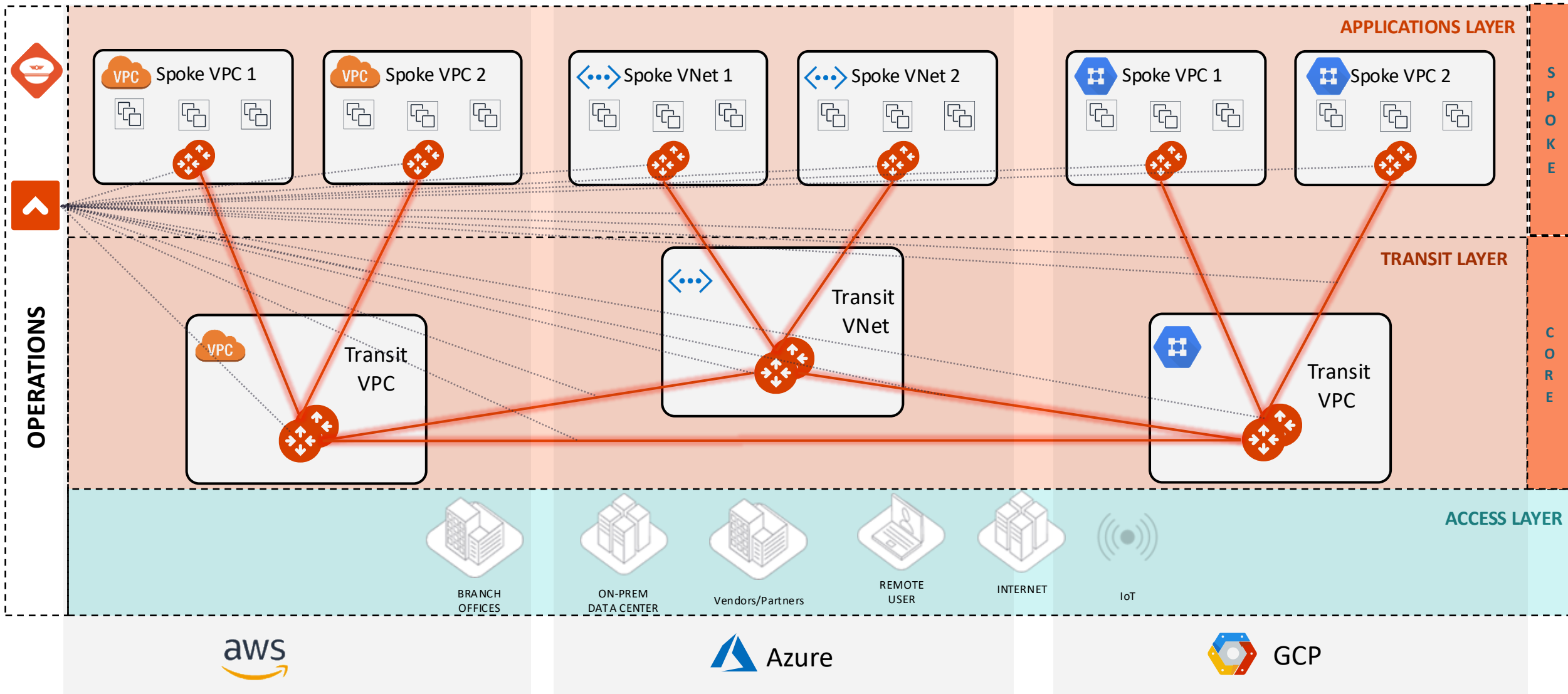
- When you identify groups of spoke and edge VPC/VNets in your infrastructure with the same requirements from a networking point of view (network reachability), you may want to group them in what Aviatrix calls “network domains”.
- A *network domain* is an Aviatrix enforced network of one or more spoke VPC/VCN/VNets.
- The key use case for building network domains is to segment traffic for an enhanced security posture. You use them, in conjunction with *connection policies*, to achieve the network isolation for inter-VPC/VNC/VNets connectivity that you want for your network.



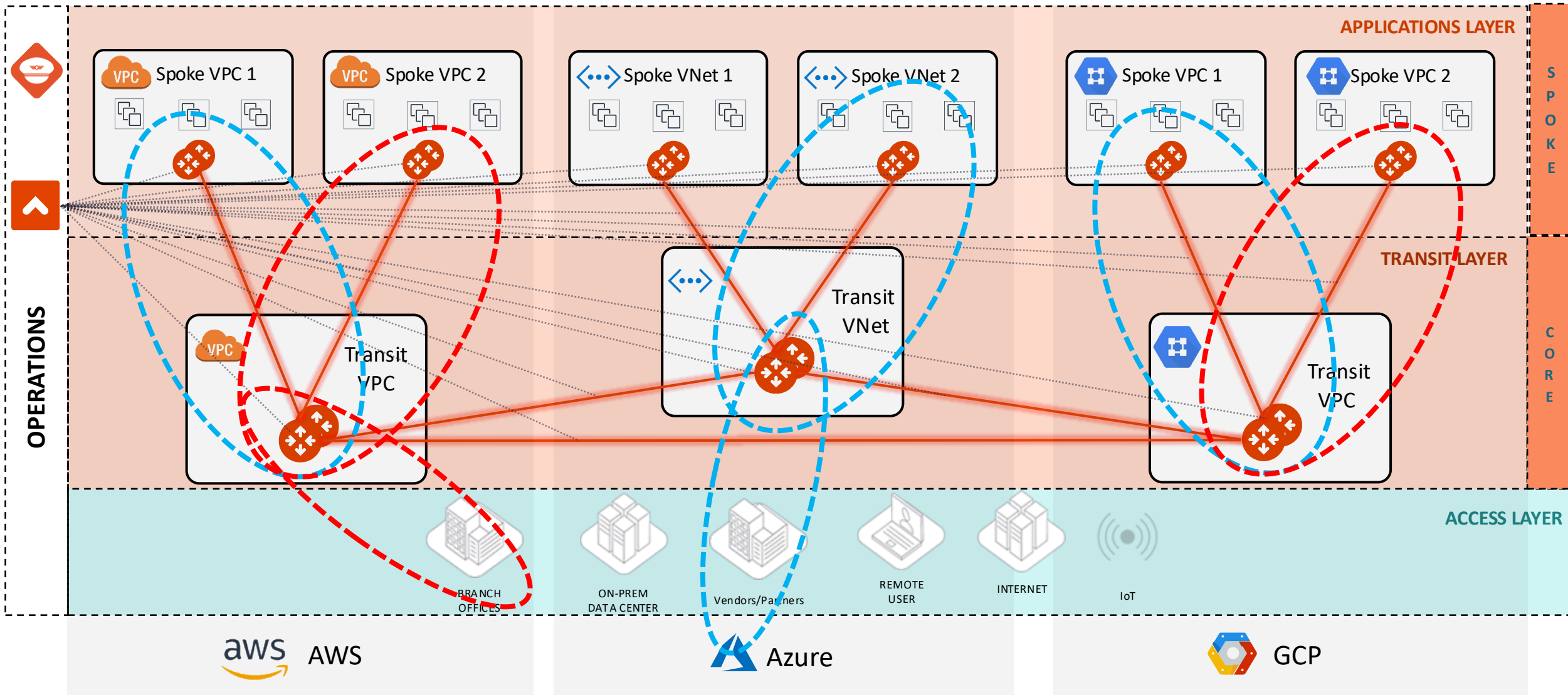
Implementing Network Segmentation in an Aviatrix-Managed Network (official documentation link):

<https://docs.aviatrix.com/copilot/latest/network-security/network-segmentation-secured.html?expand=true>

From CNSF to...



... Global Segmentation with Network Domains



Order of Operations for activating the Network Segmentation



- 1) Enable Network Segmentation on the relevant Transit Gateway(s)
- 2) Create Network Domains (aka Segments)
- 3) Associate Spoke Gateways and/or Site2Cloud connections to the Network Domains
- 4) Apply the Connection Policy (*optional*)

Gateways Overview **Transit Gateways** Spoke Gateways Specialty Gateways Gateway Management Settings

< AWS-AWS-TRANSIT-GW

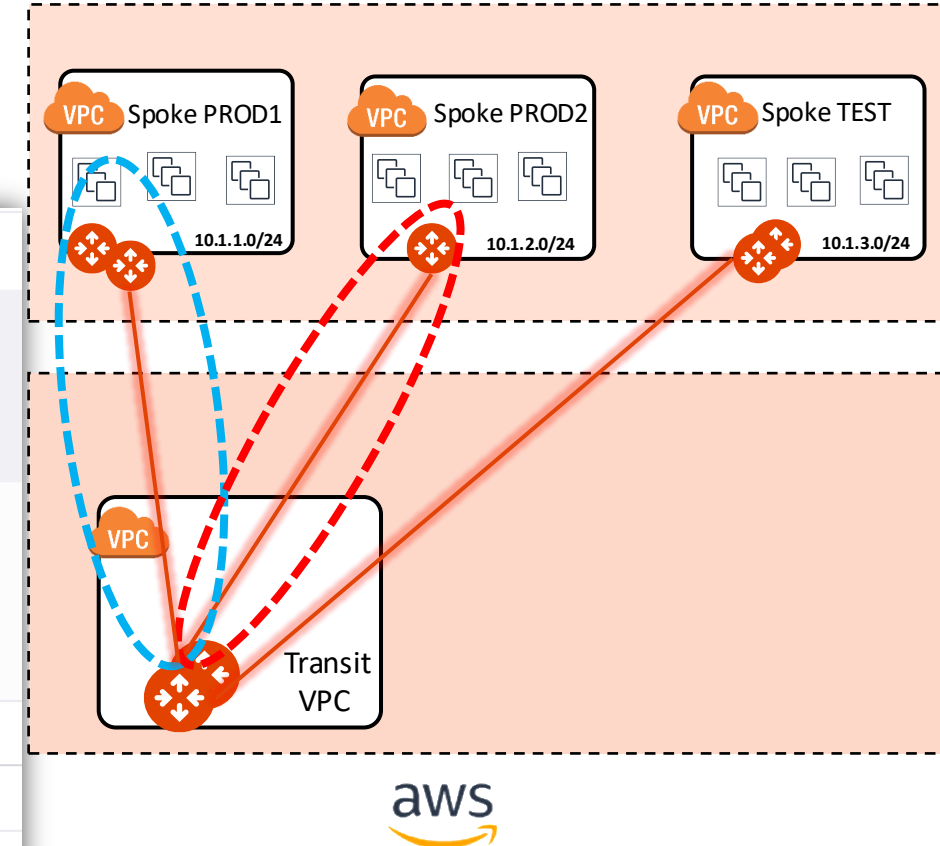
Details Instances Attachments VPC/VNet Route Tables Gateway Routes Interface Stats **Route DB** Performance Settings

Best Routes

All

Segmentation enabled

CIDR	Type	Table ID	Next Hop Gateway/Connection	Next Hop IP
10.1.1.0/24	vpc	BLUE_rtb	AVX-AWS-PROD1-GW	3.11.230.247
10.1.2.0/24	vpc	RED_rtb	AVX-AWS-PROD2-GW	18.135.173.0
10.1.3.0/24	vpc	main	AVX-AWS-TEST-GW	18.175.75.119



PATH: COPILOT > Cloud Fabric > Gateways > Transit Gateways > select the relevant GW > **Route DB** (equivalent of RIB)

Multiple Routing Domains on the Transit GW

Gateways Overview **Transit Gateways** Spoke Gateways Specialty Gateways Gateway Management Settings

< AWS-AWS-TRANSIT-GW

Details Instances Attachments VPC/VNet Route Tables **Gateway Routes** Interface Stats Route DB Performance Settings

Gateway Instance: AWS-AWS-TRANSIT-GW Network Domain: **BLUE**

Destination	Via	Interface	Next Hop IP	Next Hop Gateway	Metric
default	blackhole				400
10.1.1.0/24		tun-030BE6F7-0	3.11.230.247	AVX-AWS-PROD1-GW	100
		tun-0D2A6214-0	13.42.98.20	AVX-AWS-PROD1-GW-1	100
10.1.1.0/24		tun-0A0A0028-0	10.10.0.40	AWS-AWS-TRANSIT-GW-1	200

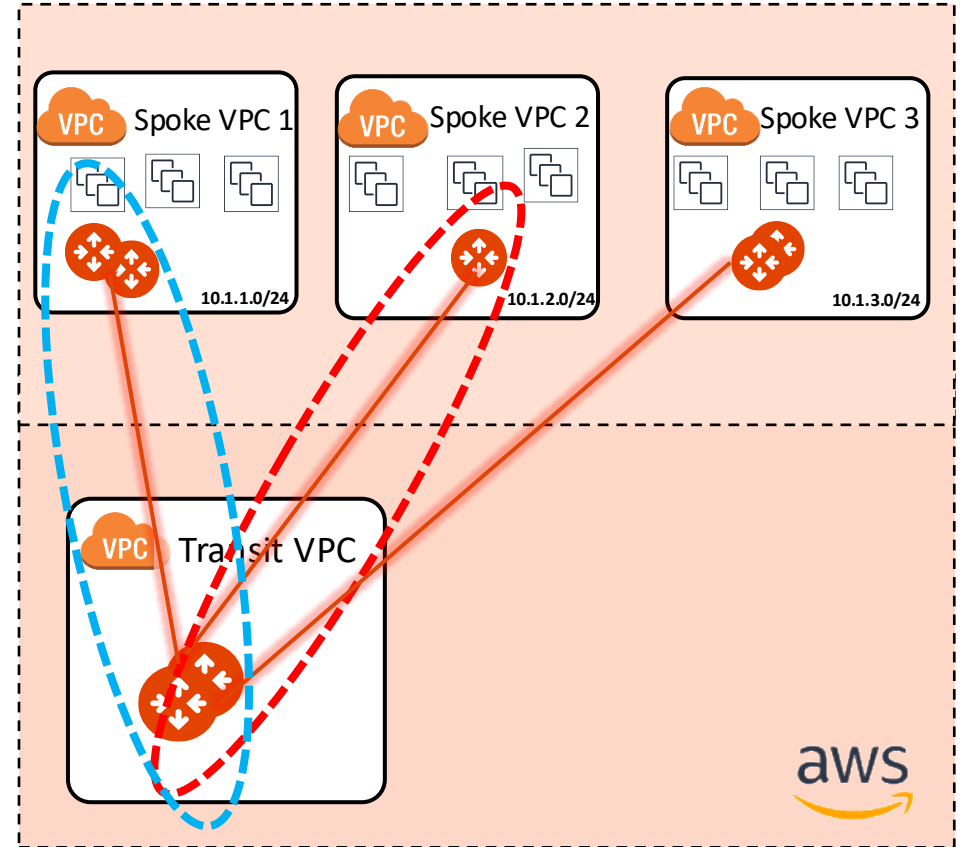
Gateways Overview **Transit Gateways** Spoke Gateways Specialty Gateways Gateway Management Settings

< AWS-AWS-TRANSIT-GW

Details Instances Attachments VPC/VNet Route Tables **Gateway Routes** Interface Stats Route DB Performance Settings

Gateway Instance: AWS-AWS-TRANSIT-GW Network Domain: **RED**

Destination	Via	Interface	Next Hop IP	Next Hop Gateway	Metric
default	blackhole				400
10.1.2.0/24		tun-1287AD00-0	18.135.173.0	AVX-AWS-PROD2-GW	100
10.1.2.0/24		tun-0A0A0028-0	10.10.0.40	AWS-AWS-TRANSIT-GW-1	200



- A single Spoke gateway or a Cluster of Spoke Gateways can be associated to a unique domain!
- **PATH:** COPILOT > Cloud Fabric > Gateways > Transit Gateways > select the relevant GW > **Gateway Routes** and then filter based on the network domain (i.e. VRF)

CAVEAT: The specific Network Domain view (aka vrf) is only available on the Transit GW. The Spoke GW has only the main routing table (aka GRT).

Connection Policy



- The Connection policy allows the **inter-domain** communication or **inter-segment** communication (sort of *vrf leaking*).
- The connection policy establishes a bidirectional connectivity (merging the network domains' RTBs).

In the example on the right, there are three domains: *Green, Blue & Yellow*

- If the Blue domain acts as the Shared Services Domain, It will be connected to both the GREEN domain and the YELLOW domain.

Edit Network Domain: BLUE

Name *

BLUE

Associations

AVX-AWS-PROD2-GW x

Connect to Network Domain

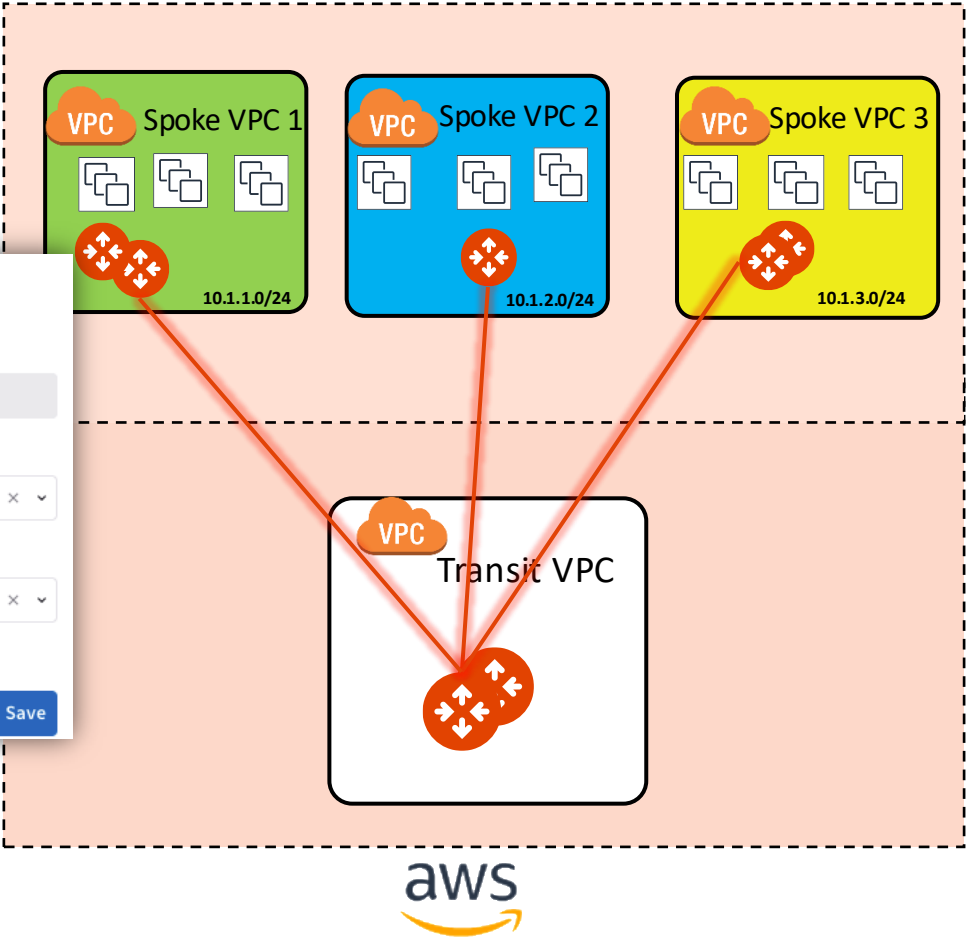
GREEN x YELLOW x

Connectivity is bidirectional

Cancel

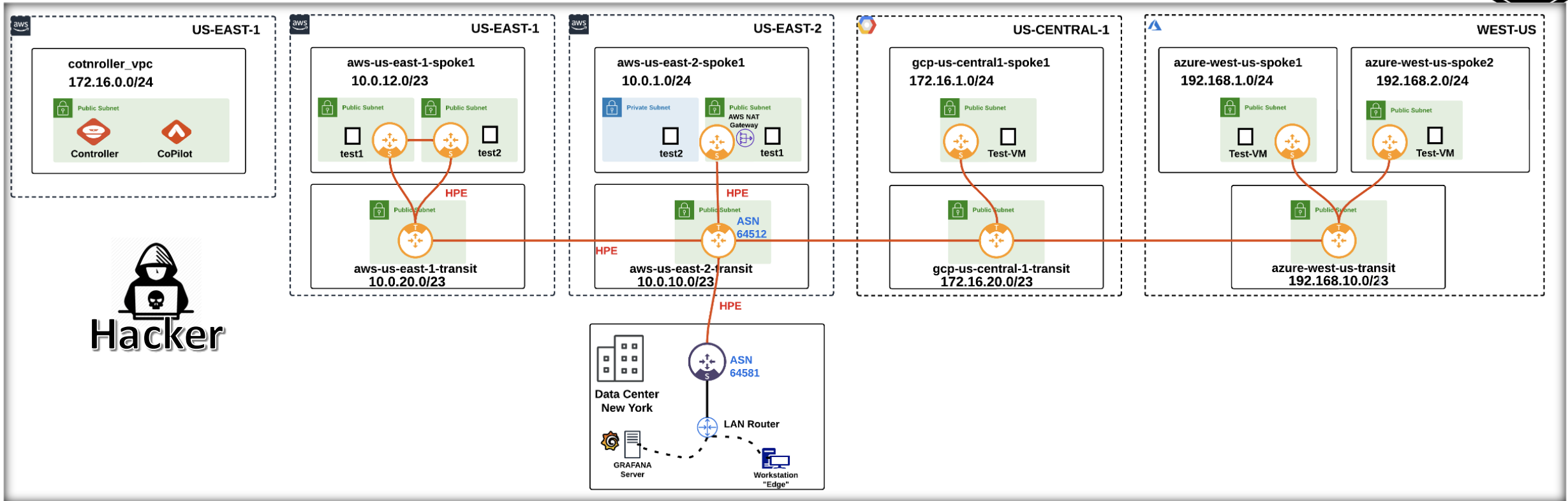
Save

Name	Associations	Connected To
YELLOW	AVX-AWS-SPOKE-GW-TEST	BLUE
GREEN	AVX-AWS-SPOKE-GW-PROD1	BLUE
BLUE	AVX-AWS-SPOKE-GW-PROD2	GREEN, YELLOW



- **CAVEAT:** a connection policy can't be applied on the main RTB (aka Global Routing Table).

Lateral Movement



- An attacker searches for an instance that could serve as a foothold for lateral
- Segmentation ensures the creation of isolated compartments.

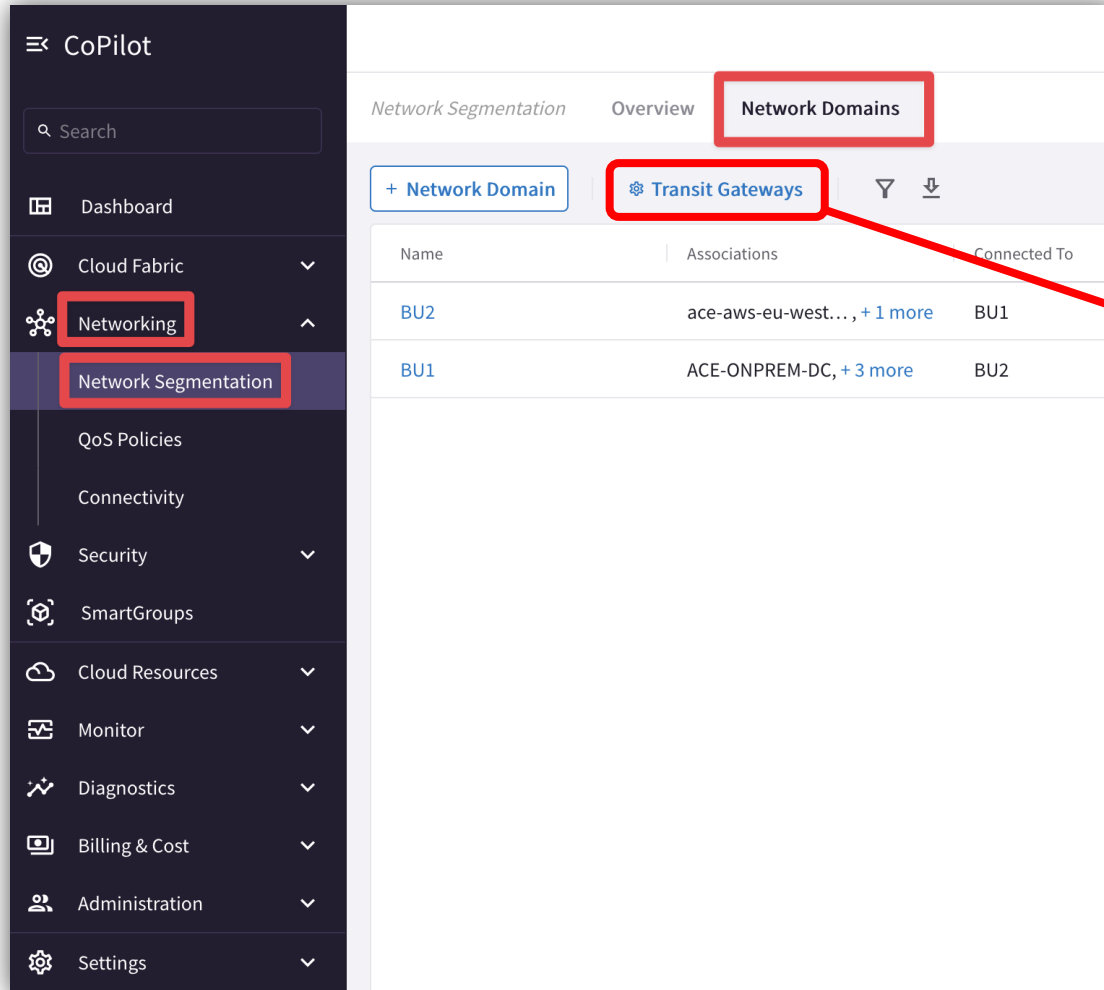


Tools for Operating Network Segmentation

Network Segmentation Visibility

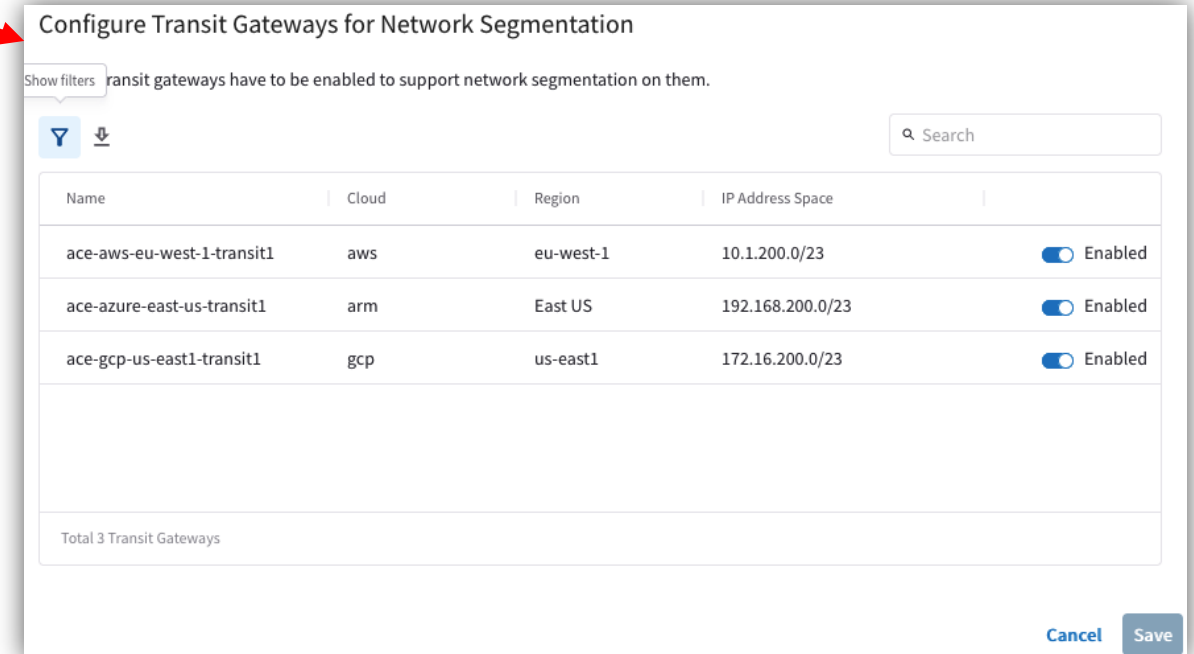
- CoPilot: verify the Network Domains

PATH: COPILOT > Networking > Network Segmentation > Network Domains



The screenshot shows the Aviatrix CoPilot interface. On the left is a dark sidebar with a search bar and a list of navigation items: Dashboard, Cloud Fabric, Networking, Network Segmentation, QoS Policies, Connectivity, Security, SmartGroups, Cloud Resources, Monitor, Diagnostics, Billing & Cost, Administration, and Settings. The 'Networking' and 'Network Segmentation' items are highlighted with red boxes. The main panel on the right has tabs for 'Network Segmentation', 'Overview', and 'Network Domains'. The 'Network Domains' tab is selected and highlighted with a red box. Below the tabs are buttons for '+ Network Domain' and 'Transit Gateways', with the latter also highlighted by a red box. A red arrow points from the 'Transit Gateways' button to a secondary window on the right.

Name	Associations	Connected To
BU2	ace-aws-eu-west..., + 1 more	BU1
BU1	ACE-ONPREM-DC, + 3 more	BU2



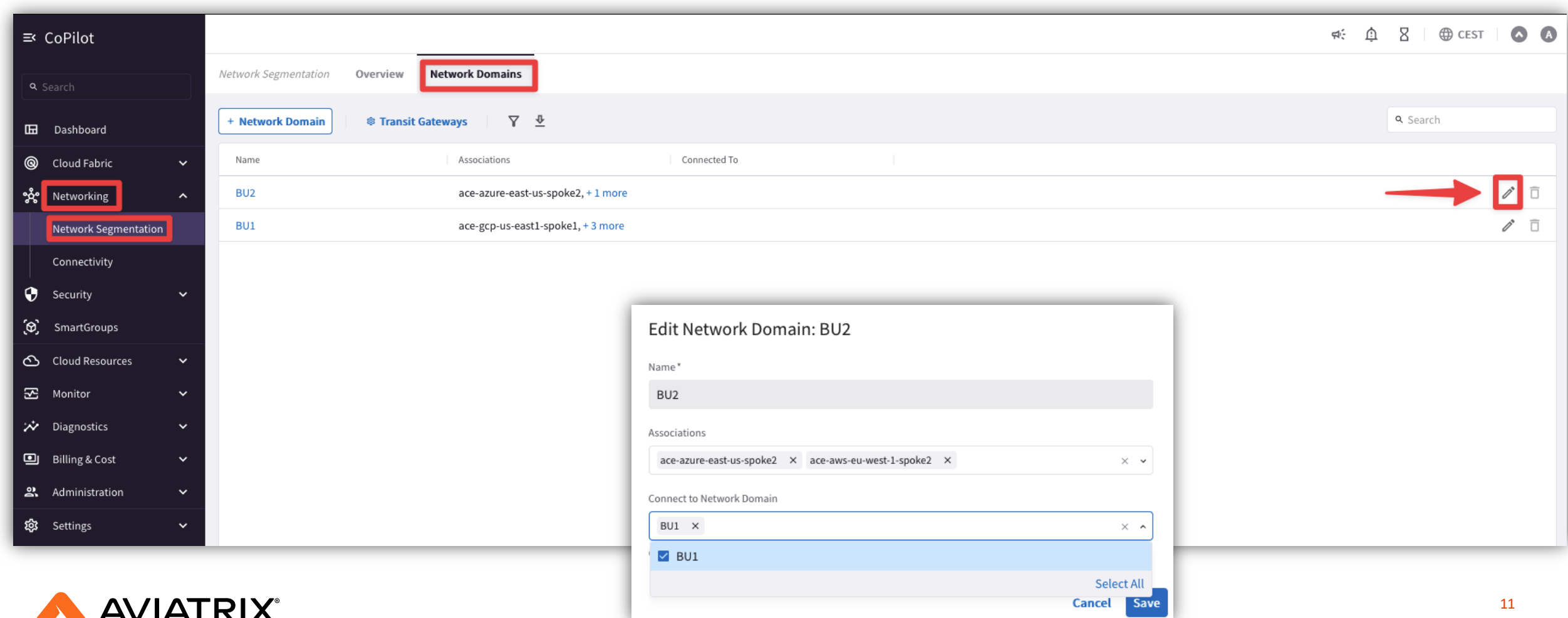
The dialog titled 'Configure Transit Gateways for Network Segmentation' displays a table of transit gateways. A message at the top states: 'transit gateways have to be enabled to support network segmentation on them.' The table has columns for Name, Cloud, Region, IP Address Space, and an enablement toggle. Three gateways are listed, all with their enablement toggles turned on. At the bottom, it says 'Total 3 Transit Gateways'. 'Cancel' and 'Save' buttons are at the bottom right.

Name	Cloud	Region	IP Address Space	Enabled
ace-aws-eu-west-1-transit1	aws	eu-west-1	10.1.200.0/23	☒ Enabled
ace-azure-east-us-transit1	arm	East US	192.168.200.0/23	☒ Enabled
ace-gcp-us-east1-transit1	gcp	us-east1	172.16.200.0/23	☒ Enabled

Network Segmentation Visibility

- CoPilot: create/modify the Network Domains

PATH: COPILOT > Networking> Network Segmentation > Network Domains > pencil icon (edit)



The screenshot displays the Aviatrix CoPilot interface. On the left, the navigation menu shows 'Networking' and 'Network Segmentation' highlighted. The main panel shows the 'Network Domains' tab with a table of domains. A red box highlights the 'Network Domains' tab, and another red box highlights the 'Networking' menu item. A red arrow points to the pencil icon in the table's action column for the 'BU2' domain. An 'Edit Network Domain: BU2' modal is open, showing the domain name 'BU2', its associations ('ace-azure-east-us-spoke2' and 'ace-aws-eu-west-1-spoke2'), and a list of connected network domains ('BU1').

Name	Associations	Connected To	
BU2	ace-azure-east-us-spoke2, + 1 more		
BU1	ace-gcp-us-east1-spoke1, + 3 more		

Edit Network Domain: BU2

Name*

BU2

Associations

ace-azure-east-us-spoke2 x ace-aws-eu-west-1-spoke2 x

Connect to Network Domain

BU1 x

☒ BU1

Select All

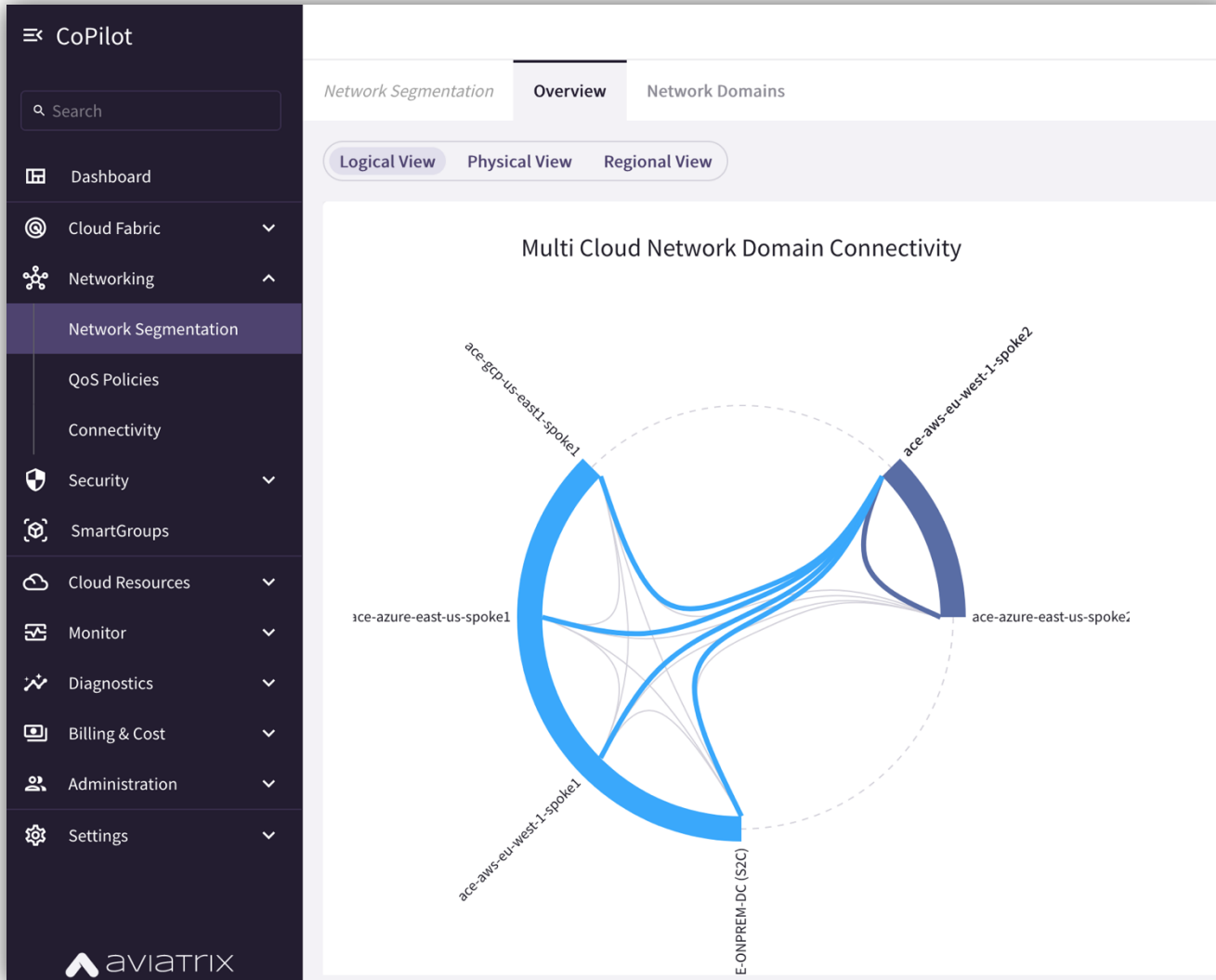
Cancel Save

Network Segmentation Visibility



- CoPilot: verify the Network Relationships

PATH: COPILOT > Networking > Network Segmentation > Overview > Logical View





Next:
Lab 1 Network Domains
&
Lab 2 Connection Policy